

Note ID: **1.2**

Master Subject: **Leviton RS232 Serial Interface RZC0P**

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Topic:

Z-Wave RS232 ASCII Interface Programming

Resolution:

1. Installation

First, include the serial interface into your Z-wave Network. Your primary remote will assign a node ID to your network.

Inclusion mode:

1. Plug RS232 module into power outlet.
2. Press and hold LED button until unit flashes Amber.
3. Select [Include Node] in your primary controller to start inclusion.

Once it has been included you will have to associate all devices you would like to control from the serial interface. From the Primary remote:

1. Select Network
2. Select HAI Association
3. Select (Contrl #) Press CENTER button
4. Place a check mark in the box next to any devices using the (+) soft key.

Note: Pressing 4ON from the primary remote will select all devices in your association list.

5. Press CENTER button to save associations
 - a. Your remote will begin to process these associations
(Note: remain in close proximity to the controller as it processes associations)

UART Settings

9600-8-N-1

2. Programming

- 2.1. Message format
- 2.2. Message start: ">"
- 2.3. Message End: "<cr>" or "<cr><lf>"
- 2.4. Input string is limited to 80 characters.

3. Commands

All commands are abbreviated to 1 or 2 characters. Below is the list of commands understood by the RZC0P serial interface.

"N"	Start association. Specifies the list of nodes IDs separated by ","
"AP"	This command will add nodes to the current association list. Example(>AP1, 103)
"GS"	This command stores the current Group represented by the association list. This parameter specifies the group number. i.e. GS1.
"GR"	Recall previously stored group. Parameter specifies the group number
"ON"	Turn ON a single light or group of lights defined by current association.
"OF"	Turn OFF a single light or group of lights defined by current association.
"L"	Sets level for single light or group of lights defined by current association. Parameter (0-99) specifies the level according to Zensys® protocol
"S"	Call scene for a single light or group of lights defined by current association. Parameters specify the following information 1: 1-255 specifies scene number, 2: specifies the fade rate (255 is default)
"PS"	Program scene for a single light or group of lights defined by current Association. Parameters specify the following information 1: 1-255 specifies scene number, 2: specifies the fade rate (255 is default)
"DI"	Start decreasing light for a single light or group of lights defined by Current association.
"BR"	Start increasing light for a single light or group of lights defined by current association.
"ST"	Stops the light change
"GR"	Recalls previously stored group. i.e GR1ON
"GS"	Stores current group
"UP"	Starts the update loop after implementing one of the commands. This command is necessary to keep all controllers in the Leviton's Z-Wave® system correctly synchronized and up to date with current network routing information.
"SE"	ends any Z-Wave message. The first parameter is the command class, the second is the command, the rest are the parameters of the appropriate command classes. Refer to the Zensys command class specification for the information on the command classes available for Z-Wave system.
"DE"	Default to reset the ASCII interface. If button on the unit is holding while the command is send the unit will reset to the factory default state.
"IN"	Switch module to the add/remove mode to add or remove serial interface to the Z-Wave network Again would we not be able to do this in the primary remote.
"AB"	Aborts current transmission
"NN"	Program/ask node name from any node in your network.
"NL"	Program/ask node location From any node in your network.
"FI"	Find a node ID with assigned properties
"RO"	Assign/delete Routes for any devices in your network Dimmer/switch/plug-ins i.e. routing slaves.

4. Command Descriptions

4.1. Start Association:

Nxxx, xxx, xxx,...xxx ,where xxx are decimal numbers representing nodes ID on a Z-wave network. To remove any association use the “N” command without parameters (use “,” if necessary to separate from the next command).

Examples:

>N1, 10,23 - Create group including 3 nodes with Ids 1,10 and 23
>N, - Create empty group, no nodes included to the group.

Question mark “?” can be used before “N” to request (BASIC GET) levels from the node specifies
>?N1,10,23 - Request node levels from nodes 1,10 and 23

4.2. Append to association:

APxxx,xxx,xxx,...xxx - Add nodes to the previously existed on the association list

?APxxx,xxx - Will request nodes’ levels from the group

4.3. Store Group

(“GS”) - Store in nonvolatile memory current association list

- N1,10,23GS1

Store group of 3 nodes 1,10,23 as a group 1.

2.14 Recall Group

(“GR”) – recall from nonvolatile memory specified group and set it as a current association list.

Example:

>GR1ON - Turn on lights in group 1

>?GR1 - Request levels for nodes in group 1.

4.4. Light Control Commands

Light control commands apply to the nodes in the current association list. If current association list is empty a broadcast message will be sent. If association list includes just one node then a single cast message will be used. If there are multiple nodes on the association list, a multicast message will be used.

All commands are using BASIC CLASS command.

Examples:

Commands bellow apply to the nodes 1, 10 and 23 (example)

>N1,10,23ON Turn ON nodes above

>N1,10,23OFF Turn OFF nodes above

>N1,10,23L50 Set nodes above to the 50% output

>N1,10,23S2,255 Call scene 2 with default fade rate for nodes above

>N1, 10, 23PS2,255 Program current levels as a scene 2 with default fade rate for nodes above.

>N,ON - broadcast ON command (all ON). Broadcast commands don’t have an acknowledgment so it is not guaranteed that all nodes will receive this message.

Light level, current scene parameters and specified scene number parameters can be asked from the nodes supporting appropriate classes.

>?Nxxx - Will request the node level/state. The response comes in form <NxxxLyyy.

>Nxxx?Sy yy - Will request scene information from a node. If yyy=0 current scene information will be returned.

Returned information is in form <NxxxS###,lll,fff, where ### scene number, lll light level for the scene and fff fade rate programmed for the scene.

4.5. Update command

After any command where changing the level or state of the load was sent, the sending node is responsible for updating information for all other controllers controlling the same nodes. To keep the system up-to-date update command should be appended to all “action” commands.

Example:

>N1,10,13ON

>UP

We would recommend a few second delay between the end of command implementation and starting the update.

4.6. Send Command (“SEnd”)

To provide control to the nodes which function a bit different than lighting control nodes the send command can be used. This command sends over RF, all Z-wave command, the node will send the message as before defined by any of the association commands:

N, AP, GR.

Example:

N5SE69,2 - Will request thermostat fan state from thermostat node 5.

4.7. NN and NL - Node name and location commands.

NN (NL) provides access to programming/recalling node name (node location name).

>NxxxNNdy yy...yyy

>NxxxNLdy yy... yyy

where xxx- node ID

yyy..yyy ASCII string up to 16 characters

To request previously programmed name use >?NN or >?NL

The name returns in format: <Nxxx”yyyyyy

4.8. FI – find node ID with requested basic, generic and specific properties.

This command allows finding network node ID for the Z-Wave node with certain basic, generic and specific classes. It maybe used during installation procedure as well to check if a node is in the routing table.

>[?]Fibbb,ggg[,sss[,lll]]

where

bbb – basic class number

ggg – generic class number

sss - specific class number

lll – instance number for the device (between 1 and 232).

If any of the class numbers (bbb,ggg,sss) is 0, the command will search for any device.

The instance number (1-232) will reference on the instance in the routing table for certain device. Basic class and generic class are mandatory fields, specific and instance fields are not. If specific field is missing it assumed 0. If instance class missing it assumed 1 (the first instance).

Zensys specifies only 4 valid basic classes:

Controller	– 1
Static controller	- 2
Slave	- 3
Routing Slave	- 4

For the generic and specific classes refer to the Zensys device class specification. For example switch device will have ggg=16, dimmer ggg=17, thermostat ggg=8.

After implementing the search found devices will be added to the current group. So you can store the current group or send any message using this group.

If '?' is set before the command interface it will report back the NodeID found as <Fxxx where xxx is a node ID for the device.

If no devices have been found to satisfy the conditions an error message will be returned
<E010

4.9. Assign routes for routing slaves.

To provide routing slaves with valid routes (when nodes can't reach each other directly) "RO" command is used.

Routing slaves can have up to 5 nodes assign return routes to them.

Because each time routes are added to the existing routing table list, we recommend deleting all routes first.

>ROxxx,0 or >ROxxx

Above command will delete all routs for the routing slave with xxx node ID.

>ROxxx,yyy

Above command will setup, maximum 4 valid routs for node xxx to node yyy.

4.10. Serial controller replies.

A Serial controller reply plays back every command send over the serial port. It also replies back after finishing every RF transmission, and when adding/removing itself to a Z-wave network. When a serial interface receives messages addressed to it, it forwards them to the serial output as well, except for messages processed by the interface itself.

4.11. Receive command replies.

After receiving commands over the serial port, the serial interface will check for any commands and then process them. If the command send was correct and the device has enough resources currently to implement the command the interface responds with : <E000. – No error message.

If any error has been detected or some of the resources requested are unavailable the node responds with <Exxx, where xxx is the error number. Possible errors are listed bellow:

0 - no error

1 - wrong start of the string symbol

2 – Input buffer overflow.

3 – Can not start RF transmission. All buffers are taken.

4 – Can not start RF transmission because previous one has not finished.

5 –Unrecognized command

6 – Attempt to send the new buffer over RS232 before previous one had been processed.

7 – The send Message does not have data fields specified.

8 – Can not stop SUC mode. Node is SUC.

- 9 – EERPOM is busy, can't store group information
- 10- No devices with specified properties have been found.

The new input string can be accepted only after response <Exxx was received as a result of processing the previous one. Attempting to send a new input string before previous one has been processed will cause an Error#6 code.

4.12. Replies after implementing RF transmission commands.

After implementing the transmission requested the serial interface will reply with the string:

<Xyyy, where yyy is the error code.

yyy=0 if transmission was successful or yyy=2 if an error was detected during transmission.

After receiving <Xyyy the next command transmission can be send.

4.13. Add/remove node replies.

While going through the stages of add/remove to the Z-wave network the interface will reply with the strings:

<!xxx,

Where xxx=16 if Learning mode has started and xxx=32 if this mode is unsuccessful, and xxx=0 if successful.

4.14. Received message strings.

If BASIC REPORT, MULTILEVEL_SWITCH or BINARY_SWITCH reports are received the module will transfer it to the serial port as

<NxxxLyyy

Where xxx is a node ID and yyy is a light level between 1and 100. 0 indicates off and 255- On.

The scene information reported back as

<NxxxS###,lll,fff

where ### scene number, lll light level, fff fade rate.

Scene number 0 indicates that the node is not at any scene level.

Node name information displays in format:

<Nxxx"yyy..yyy for the node name and <Nxxx'yyy...yyy for node location name.

All other general messages are displayed in general format:

<Nxxx:yyy,yyy,.....

where xxx is a node ID and yyy decimal command data.

The first byte represent command class, the second command and the rest of the bytes represent the command data. Refer to the Zensys Command class specification for information regarding the data structure for the different commands.

4.15. Found node ID reply.

If a node with specified properties has been found in the routing table and '?' is present in front of the request, the node id is returned as a following message:

<Fxxx ,

Where xxx is the node ID for the node been found.

1. Appendix.

A. Multilevel Sensor.

Below is an example of getting information from Multilevel Sensor (Temperature sensor used by thermostats). Assumed that sensor node ID is 4.

```
>N4SE49,4          <- Send multilevel sensor GET
<E000              <- RS interface processed the input
<X000              <- Appropriate message has been send correctly
<N004:049,005,001,009,075  <- REPORT from the sensor:
```

049 – Command class
005 – Command report
001 – The value send is temperature
009 – The value represented by 1 byte and in deg. F
075 – Temperature 75F

B. Control thermostat fan.

Assume that thermostat node is 4.

```
>N4SE68,1,1          <- Set thermostat fan mode 1: Will turn the manual fan operation on.

<E000                <- RS interface processed the input
<X000                <- Appropriate message has been send correctly
>SE68,2              <- request thermostat fan mode (current group is still just node 4, so you do not need to
repeat N4 command.
<E000                <- RS interface processed the input
<X000                <- Appropriate message has been send correctly
<N004:068,003,001    <- Thermostat report indicate mode 1
>SE68,1,0            <- Set thermostat fan mode 0: Will turn the manual fan operation off .

<E000                <- RS interface processed the input
<X000                <- Appropriate message has been send correctly
>SE68,2              <- Request thermostat fan mode
<E000                <- RS interface processed the input
<X000                <- Appropriate message has been send correctly
<N004:068,003,000    <- Thermostat report indicate mode 1
```

C. Set Thermostat Set point.

Assume again thermostat is node 4.

```
>N4SE67,1,2,9,80      <- Set thermostat cooling set point to 80F
<E000
<X000
>N4SE67,2,2          <- Request thermostat set point information
<E000
<X000
<N004:067,003,002,009,080  <- Get back thermostat report with set point 80F.
```

D. Dimmer control

Assume dimmer node is node 8.

```
>?N8          <- request dimmer level
<E000         <- request processed
<X000         <- request has been sent
<N008L100     <- received dimmer's response, current level 100%

>L50UP        <- Set light level at 50% and request update
<E000         <- request processed
<X000         <- request has been sent
<N008L050     <- Node 8 sends back it current level as result of update. Nothing will be
send back if update wasn't requested.
```

Turn dimmer OFF sequence.

```
>N8OFF,UP
<E000
<X000
<N008L000
```

E. Updates with few nodes

```
>?N8,4        <- Request state of 2 nodes (8 –dimmer, 4- thermostat)
<E000         <- request processed
<N008L050     <- response from dimmer
<N004:064,003,003 <-response from thermostat
<X000         <- requested RF transmission (GET) is finished
<N008L050     <- response from thermostat
>N8,4UP       <- Instead of requesting response use update
<E000         <- request processed
<N004:064,003,003 <-response from thermostat
<N008L050     <- response from dimmer
```

F. If node ID for whatever reason is unknown the find node ID command can be used.

```
>N,           <- empty current group
>FI0,8 SE67,1,2,9,80 <- Set the first thermostat cooling set point to 80F
<E000
<X000
>GS100        <- Store the thermostat as a group 100 (to recall this node instead of
searching every time)
Find out what thermostat node ID:
>?FI0,8
<F004
<E000
```

Interface reported that the first found thermostat has nodeID=4.

D. Setup Association tables for node.

To setup association table for the node use association class.
SE133,1, Group#, node1,node2,...

Where Group# is 1 for the dimmers/switches (since they support just one group) and 1 through 4 for 4

buttons controller based n what button you are programming.
node1,node2 are the nodes ADDED to the node list for the dimmer or controller's button
To remove node use
SE133,4, Group#, node1,node2,...

For example:

>N2SE133,1,1,7

<E000

<X000

The above will add node 7 to the association table for the dimmer.

Revisions

- 1.0 Original
- 1.1 New format, document renamed
- 1.2 Renumbered

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